Overcoming Visual Clutter in Vision Language Action Models via Concept-Gated Visual Distillation

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Abstract*— Vision-Language-Action (VLA) models demonstrate impressive zero-shot generalization but degrade sharply in cluttered scenes, a failure we term the "Precision-Reasoning Gap" and hypothesize to reflect background-induced feature dilution: semantically confusable clutter corrupting the visual grounding required for precise manipulation. We propose Concept-Gated Visual Distillation (CGVD), a training-free, model-agnostic inference-time framework that parses the instruction into a safe set, applies a supplied distractor vocabulary with a two-layer target refinement (cross-validation, spatial disambiguation) that penalizes false positives, and inpaints the remaining distractor regions, preserving scene context outside the edits. Compared against the un-augmented π_0 and GR00T policies in SimplerEnv, CGVD substantially improves robustness to semantic clutter—on *put spoon on towel* with 18 semantic distractors, π_0's success rate rises from 43.0% to 77.5% (Table III)—and, in one exploratory 5-seed condition with compositional prompts, improves attribute adherence. At the same condition, a vocabulary-matched reimplementation of the run-time intervention baseline BYOVLA reaches 32.5% at $\approx 30\times$ CGVD's total per-chunk cost—a dense-clutter regime outside BYOVLA's design envelope (§IV-E). The protection is task- and distractor-type-conditional: on *put carrot on plate*, where contextual clutter aids the baseline policies, CGVD provides no benefit at any tested density and often significantly underperforms them. Runtime compositing adds roughly 16 ms of per-frame overhead after a sub-second per-episode initialization. These results support inference-time visual distillation as a practical defense against semantically confusable clutter for frozen VLA policies in static scenes with a known clutter vocabulary.

I. INTRODUCTION

Vision-Language-Action (VLA) models [1]–[5] ground large language models into robotic control, enabling robots to follow open-vocabulary instructions like "Put spoon on towel" without task-specific training. These models promise a future where robots can operate in unstructured, human-centric environments.

However, a significant gap remains between the semantic reasoning capabilities of these models and their geometric precision in deployment conditions. While VLAs excel in curated, clutter-free environments, their performance degrades precipitously in the presence of visual clutter [6], and more broadly under physical variations of the scene [7]. We term this phenomenon the Precision-Reasoning Gap: the model successfully identifies the target object conceptually, yet fails to manipulate it precisely. We hypothesize that attention corruption from surrounding distractors dilutes the latent representation used for spatial planning; under this hypothesis, feature dilution would manifest as high-variance trajectories and manipulation failure. Our experiments (§IV) measure the end-to-end effect of removing distractors; they do not directly instrument the attention mechanism, so we present this account as a working hypothesis rather than an established mechanism.

This degradation is not uniform across distractor types: failure concentrates around distractors sharing visual or semantic properties with the target, consistent with prior reports [6]. While VLAs may exhibit resilience to arbitrary clutter via large-scale pre-training, they remain brittle to semantically confusable objects; for example, a fork scattered near a target spoon triggers conflicting visual tokens within the same affordance category, causing the policy to attend to or even grasp the wrong object.

Existing mitigations fall into three categories (§II): observation-grounding methods such as OBEYED-VLA [8] augment the VLA with a trained perception module that produces task-conditioned, object-centric, geometry-grounded observations, fine-tuning the policy on clean demonstrations; inference-time interventions such as BYOVLA [9] identify distractors with a VLM (an external GPT-4o dependency) and probe per-region sensitivity with multiple VLA forward passes, with threshold-based target protection; and training-time augmentation [10]–[12] generates cluttered training data at the cost of retraining, without deployment-time guarantees.

To bridge this gap without retraining, we propose Concept-Gated Visual Distillation (CGVD): a model-agnostic inference framework leveraging vision foundation models [13] to restructure observations before they reach the policy. CGVD parses the instruction into target and anchor, segments distractors and task-relevant entities independently, and uses set-theoretic subtraction to exclude the target from the distractor mask *conditional on correct safe-set segmentation*: no pixel segmented as safe is inpainted, but an object the segmenter misses is unprotected (§IV-D). Distractors are then replaced via inpainting [14], preserving the surrounding scene context.

Our contributions are as follows:

- **Concept-Gated Visual Distillation (CGVD):** a training-free, model-agnostic inference framework removing distractors from VLA observations via language-grounded segmentation and inpainting while preserving scene context.
- **Interaction-Aware Masking Logic:** a set-theoretic cross-validation pipeline that overcomes the semantic confusion of open-set segmenters (which score prompts

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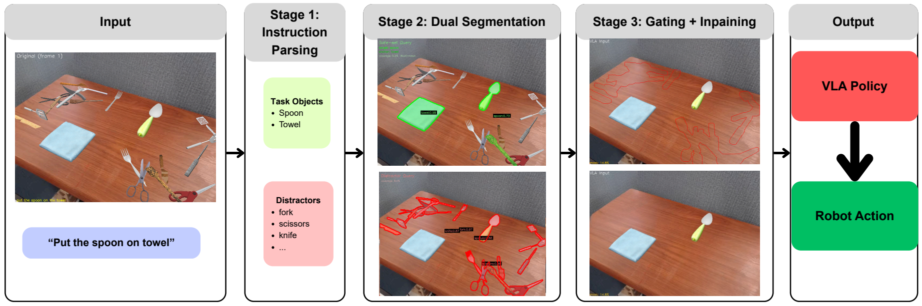

Fig. 1. The CGVD pipeline. **Stage 1:** the instruction is parsed into a safe set (target, anchor); the distractor vocabulary is supplied. **Stage 2:** SAM 3 segments both sets independently. **Stage 3:** set-theoretic gating subtracts the safe-set mask from the distractor mask; LaMa inpaints the result into the distilled observation passed to the VLA.

independently), penalizing false positives and spatially disambiguating true targets from confusable distractors.

- **Systematic clutter-density evaluation:** We evaluate CGVD on two VLAs (π_0, GR00T) within the SimplerEnv benchmark over more than 20,000 episodes, comparing each policy with and without CGVD under matched seeds. CGVD prevents policy collapse under dense *semantic* clutter—reaching 77.5% success against the un-augmented policy's 43.0% (π_0, *put spoon on towel*, 18 semantic distractors; Table III)—and improves attribute adherence in an exploratory compositional-prompt study, while *underperforming* the baseline where contextual clutter aids the policy (§IV-B). Comparisons are against the un-augmented policies, plus a measured head-to-head against a BYOVLA [9] reimplementation (§IV-E).

II. Related Work

A. Vision-Language-Action Models

VLA models such as RT-2 [2], OpenVLA [1], and Octo [3] leverage internet-scale pre-training for zero-shot generalization, yet struggle with high-precision geometric grounding in clutter [6]. We hypothesize that the static patch resolution of ViT backbones [15] and the absence of coarse-to-fine zooming [16] make such architectures susceptible to feature dilution [6]: background clutter competing with task-relevant signals for representational capacity.

B. Vision Foundation Models for Robotic Perception

Vision Foundation Models such as SAM 3 [13] and GroundingDINO [17] localize objects from free-form text; robotics has mostly used them to *add* information—value maps [18], semantic maps [19], labeling [20], visual prompting [21]. Our work inverts this paradigm, leveraging the open-set discriminative power of VFMs to suppress task-irrelevant regions instead; the resulting mask is an information bottleneck only in an informal, analogical sense [22], with no learned rate–distortion trade-off.

C. Robustness via Training-Time Augmentation

Domain randomization [23] and large-scale pre-training [24], [25] have not made VLAs robust to cluttered-scene shifts [26], [27]. Training-time augmentation generates cluttered or counterfactual data: ROSIE [10], GenAug [11], and NICE [12], whose pixel-space *scene surgery* is the closest prior to our core operation. CGVD performs a related pixel-space edit at *inference* time on a frozen policy: no retraining, instruction-adaptive, but inheriting run-time perception risk.

D. Inference-Time Attention Intervention

Distracting Token Pruning (DTP) [28] suppresses text-misaligned visual tokens in feature space; we conjecture such pruning struggles once distractor and target tokens are entangled by early self-attention, and instead intervene in pixel space, before tokenization. Unlike Visual Prompting [21], which adds information, we remove it. Two prior methods also operate on the observation: OBEYED-VLA [8] pairs the VLA with a *trained* perception module (task-conditioned, object-centric, geometry-grounded views) and fine-tunes the policy on clean single-object demonstrations, whereas CGVD trains and fine-tunes nothing. BYOVLA [9] likewise intervenes at run time, and §IV-E reports a measured comparison.

III. Methodology

CGVD is a training-free, inference-time perception wrapper around any VLA policy (Fig. 1): given instruction l and observation $o_t \in \mathbb{R}^{H\times W\times 3}$, a distillation function ϕ produces $\hat{o}_t = \phi(o_t, l)$ with distractors replaced by background, and the policy acts on $a_t = \pi(\hat{o}_t, l)$. Since ϕ touches only the observation interface, it is architecture-agnostic. The key insight is that the instruction already specifies which objects must remain visible, and CGVD uses this signal to gate visual content; the clutter vocabulary is supplied separately (§III-A).

A. Concept-Gated Decomposition

CGVD begins by parsing the language instruction l to extract a target concept c_{tgt} and an anchor concept c_{anc}. For instance, "put spoon on towel" yields $c_{\text{tgt}} = \texttt{spoon}$ and $c_{\text{anc}} = \texttt{towel}$. These concepts define two sets:

- The **safe set** $\mathscr{S} = \{c_{\text{tgt}}, c_{\text{anc}}\}$: entities that must remain visible. The robot arm is also always preserved, but it is handled through a separate proprioceptive channel (§III-E–III-F) rather than through the segmentation-based safe-set mask.

- The **distractor set** $\mathscr{D} = \{d_1, \ldots, d_K\}$: semantic categories to be treated as clutter (e.g., spatula, fork, knife).

The safe set is derived deterministically from the instruction; the distractor set, by contrast, is an externally supplied vocabulary. In our experiments, $\mathscr{D}$ is re-derived at every episode reset from the category labels of the objects the evaluation protocol actually spawned in that episode (§IV-A)—i.e., CGVD receives the exact ground-truth clutter inventory, per episode. At deployment, $\mathscr{D}$ would have to be supplied by the integrator as a site-specific vocabulary or proposed by an external model. Where prior work uses an LLM to determine relevance [9], our parsing is deterministic and needs no API *in this closed-vocabulary setting*—a different point on a coverage-versus-cost trade-off, not a strict improvement. §IV-D measures this dependence directly at the tested operating point.

B. Text-Prompted Instance Segmentation

Both the safe set and distractor set are segmented from the observation using SAM 3 [13]. The two sets produce independent mask channels:

$$M_{\text{dist}} = \bigcup_{d_k \in \mathscr{D}} \textsc{Seg}(o_t, d_k), \tag{1}$$

$$M_{\text{safe}} = \textsc{Seg}(o_t, c_{\text{tgt}}) \cup \textsc{Seg}(o_t, c_{\text{anc}}), \tag{2}$$

where $\textsc{Seg}(o_t, c)$ denotes the union of all instance masks returned by SAM 3 for concept c in observation o_t. As a computational optimization, the vision encoder is executed only once, on the initialization frame, with its masks reused across the episode.

C. Two-Layer Target Refinement

Open-set segmenters evaluate text prompts independently, so single-pass detection suffers semantic confusion: a spatula can yield high confidence for "spoon." To ensure the true target survives distillation, we apply a two-layer refinement on the initial frame (t=0).

Layer 1: Cross-Validation. Each target instance s_i receives a genuineness score—the confidence differential between its safe-set and distractor identities:

$$g(s_i) = \sigma_{\text{safe}}(s_i) - \max_{\substack{d_j \in \mathscr{D} \\ \text{IoU}(s_i, d_j) > \eta}} \sigma_{\text{dist}}(d_j), \tag{3}$$

where σ_{safe} and σ_{dist} are object confidences, and η is an IoU threshold ($\eta = 0.30$). When no distractor satisfies $\text{IoU}(s_i, d_j) > \eta$—the common case for an isolated target—the maximum is defined as 0, so $g(s_i)$ reduces to $\sigma_{\text{safe}}(s_i)$. Genuine targets yield $g > 0$, while imposters yield $g < 0$. Negative values are explicitly preserved to actively penalize false positives in the next layer.

Layer 2: Spatial Disambiguation. Even after cross-validation, the target mask may contain fragmented artifacts or multiple disjoint physical objects. To isolate the correct entity, we evaluate each connected component C_k using a composite score:

$$\text{score}(C_k) = \big(1 + g^*(C_k)\big) \cdot \sigma^*(C_k). \tag{4}$$

Here, $g^*(C_k)$ is maximum genuineness, and $\sigma^*(C_k)$ is peak safe-set confidence. These factors jointly favor genuine and high-confidence components. Only the top-scoring component is retained; the pipeline therefore assumes a *single* target instance per scene (§V).

D. Concept-Gated Mask Composition

The refined masks are combined into a final inpainting mask via set-theoretic gating:

$$M_{\text{inp}} = \text{dilate}(M_{\text{dist}}, r_d) \setminus \text{dilate}(M_{\text{safe}}, r_s), \tag{5}$$

where r_d is a distractor dilation radius, and $r_s \geq r_d$ is a safe-set dilation radius that creates a protective buffer; the implementation enforces $r_s \geq r_d$ by construction, and both are 11 px in our configuration. All masks are binarized with a threshold of 0.5 before dilation to eliminate soft-value artifacts.

E. Clean Scene Generation via Inpainting

We generate a single clean scene—an image of the workspace with distractors removed—by applying LaMa [14], a Fourier convolution-based inpainting model, to the initial frame. The inpainting mask is constructed as:

$$M_{\text{lama}} = M_{\text{inp}} \cup \text{dilate}(M_{\text{robot}}, r_e), \tag{6}$$

where M_{robot} is the robot mask (obtained as described in §III-F) and r_e is a robot dilation radius (20 px). Inpainting the robot region synthesizes the background *behind* the arm, leaving no stale robot pixels once it moves. A consequence is that scene content occluded by the arm at t=0 is synthesized rather than observed for the remainder of the episode (§V). LaMa fills the masked regions with photorealistic background texture. The clean scene is computed once per episode and cached for all subsequent frames.

F. Temporally Consistent Compositing

At each timestep $t > 0$, the distilled observation is produced by alpha-blending the live camera frame with the cached clean scene:

$$\hat{o}_t = \alpha \odot \hat{o}_{\text{clean}} + (1 - \alpha) \odot o_t, \tag{7}$$

where the compositing mask $\alpha \in [0,1]^{H \times W}$ is computed *once* at t=0 and held fixed: the binarized, safe-set-gated distractor mask is feathered with a Gaussian blur ($\sigma = 3.0$), clamped to 1 in core distractor regions and 0 in the (dilated) target region. So that inpainting artifacts never obscure the arm, robot pixels are overwritten onto the composite every step—the only per-frame operation. In simulation this robot mask comes from SimplerEnv's renderer segmentation buffer (per-link actor IDs): pixel-perfect, jitter-free, and *simulation-privileged*; §IV-D measures the cost of substituting a per-frame SAM 3 prediction; on hardware, forward-kinematics projection provides a deterministic, perception-free replacement.

Two properties of this design should be stated explicitly. First, $\hat{o}_t$ is the only visual input the policy receives. Second, the characteristic failure mode of observation editing is *false-positive erasure*: an object misclassified as a distractor—or entering the workspace after $t=0$, where the fixed α can blend it toward the cached background—is removed from the policy's view while remaining in the world, and the arm can collide with what it cannot see. Our evaluation is therefore scoped to static workspaces with no human entry, a fixed camera, and a known clutter vocabulary.

IV. Experiments

We evaluate CGVD across two VLA architectures on tabletop manipulation tasks with controlled distractor injection. Our experiments address three questions: (1) Does CGVD improve clutter robustness across different VLA architectures and tasks? (2) How does performance scale with distractor density? (3) How do individual CGVD components contribute at the operating point tested?

A. Experimental Setup

Environment. We evaluate in SimplerEnv [29], whose real-to-sim correlation supports *policy-level* transfer. It does not bound CGVD's perception-level risk: SAM 3 [13] and LaMa [14] were trained on real imagery and are applied here to rendered frames—itself a domain shift—and two pipeline inputs are simulation-privileged (the renderer robot mask, §III-F; the distractor vocabulary, §III-A). All conclusions are scoped to simulation (§V).

All experiments use the WidowX robotic arm with a single fixed third-person camera, matching the Bridge dataset training setup. We test two Bridge dataset tasks: (1) *put spoon on towel* and (2) *put carrot on plate*.

Policies. π_0 refers to the open-source `open-pi-zero` reproduction of π_0 [5] (`github.com/allenzren/open-pi-zero`), Bridge checkpoint `bridge_beta`; GR00T refers to `nvidia/GR00T-N1.6-bridge` [4].[1] Both are frozen; CGVD touches only their image input.

Distractors. We introduce a three-way distractor taxonomy: *Semantic* (high semantic proximity and shared visual characteristics with the target), *Random* (no similarity), and *Attribute* (same category and form, different physical properties); [6] documents the magnitude of clutter-induced degradation but uses a continuous clutter measure rather than these types. Assets come from RoboCasa [27] and YCB [30], placed collision-aware on a fixed 6 cm grid (one object per cell); an unplaceable object falls back to a collision-free random position or is hidden off-scene—episodes are never dropped. The fallback engages at the highest density: across 1,400 audited re-executions of the headline condition (identical placement generator and seeds), 18.7% of episodes realized fewer than the nominal 18 objects (mean 17.6), which makes realized-density trends, if anything, conservative. The vocabulary $\mathscr{D}$ is re-derived per episode from the actually-spawned assets' category labels (`rc_fork_0` → "fork"): it coincides exactly with each episode's injected inventory (§III-A).

Protocol. We report success rate (SR) via SimplerEnv's built-in per-task predicate over 60-step episodes: 20 episodes × 10 seeds per condition (200 episodes; the attribute study of Table II uses 5 seeds), with matched seeds between arms. Seed $s \in \{0,\ldots,9\}$ seeds the RNGs; episode e uses canonical initial layout $(s+e)$ mod 24 and distractor seed $10^4 s+e$, so both arms see identical states and clutter. Counts are sampled at $\{0,4,8,10,14,18\}$; no episodes were excluded. We compare CGVD against the un-augmented policies throughout, and against a reimplementation of BYOVLA [9] at the headline condition (§IV-E). OBEYED-VLA's released code covers its perception module but not the action policy or robot interface, so an end-to-end comparison was infeasible; DTP's [28] only advertised release is an anonymized review-time repository that has since expired.

Analysis. We report episode-level mean success rates with, as dispersion, the SD of the 10 per-seed rates; differences carry 95% CIs from the per-seed paired deltas (t_9). Episode-level clustering by seed is negligible (one-way ICC across all arms: median -0.02, max 0.06). The headline contrast—π_0, *put spoon on towel*, 18 semantic distractors—is the single confirmatory comparison ($\Delta = +34.5$ pp, 95% CI $[+20.8, +48.2]$, paired t $p = 3\times10^{-4}$; Wilcoxon $p = 0.002$); all other contrasts are exploratory and unadjusted, and deltas whose CI contains zero are within run-to-run noise. The 0-distractor cells are shared between the semantic and random conditions (physically identical). The one condition executed twice end-to-end (π_0, *carrot*, random, 18 distractors) differed between batches by at most 3.0 pp, a direct run-to-run replication estimate.

Implementation. All CGVD parameters, held fixed across every reported experiment: SAM 3 checkpoint `facebook/sam3`; LaMa big-lama (`simple-lama-inpainting`); confidence thresholds 0.30 (safe set) and 0.20 (distractors); $\eta = 0.30$; $r_d = r_s = 11$ px; $r_e = 20$ px; compositing $\sigma = 3.0$; mask binarization 0.5; minimum connected component 50 px; one warm-up frame; clean scene cached once per episode, never refreshed; SAM 3 prompts are the instruction's noun phrases and the category labels of $\mathscr{D}$. Values were selected on small pilot rollouts (primarily spoon-task) and frozen before all reported runs (verified from version-control history). Code, configuration, and evaluation scripts are available at `github.com/awd1779/pi0`.

B. Main Results

Figure 2 plots success rate against distractor count (0–18); Table I reports the numbers with per-seed dispersion and paired CIs.

First, in the presence of semantically confusing clutter, baseline performance degrades precipitously, and applying

[1] Ref. [4] documents GR00T N1; the evaluated checkpoint is the N1.6 revision (different backbone and action head) — see the NVIDIA model card.

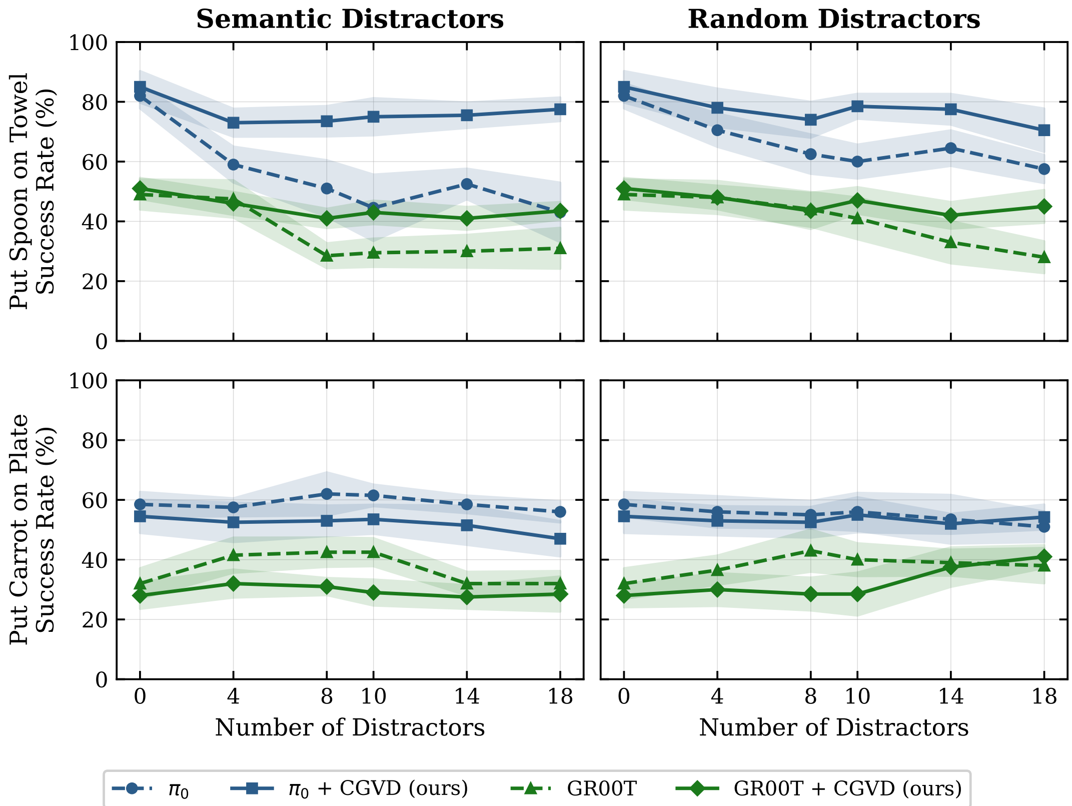

Fig. 2. Success rate vs. distractor count. **Left:** semantic; **right:** random distractors. **Top:** *put spoon on towel*; **bottom:** *put carrot on plate*. Dashed: baseline VLA; solid: +CGVD; colors denote architectures. Each point is the mean over 200 rollouts (20 episodes $\times$ 10 matched seeds) at counts $\{0,4,8,10,14,18\}$ (19,200 episodes total); 0-distractor points are shared between panels. Numeric values with dispersion and CIs: Table I.

TABLE I

MAIN RESULTS (NUMERIC). SUCCESS RATE (%), MEAN $\pm$ SD OVER 10 MATCHED SEEDS (200 EPISODES/CELL); Δ WITH 95% CI FROM PER-SEED PAIRED DELTAS (t_9), BOLD WHEN THE CI EXCLUDES ZERO. DATA OF FIG. 2.

		Semantic distractors						Random distractors					
		π_0			GR00T			π_0			GR00T		
Task	**#D**	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]
spoon on towel	0	82.0	85.0	+3.0 [-5.6,+11.6]	49.0	51.0	+2.0 [-6.1,+10.1]	82.0	85.0	+3.0 [-5.6,+11.6]	49.0	51.0	+2.0 [-6.1,+10.1]
	4	59.0	73.0	**+14.0 [+5.4,+22.6]**	47.5	46.0	-1.5 [-8.7,+5.7]	70.5	78.0	+7.5 [-0.8,+15.8]	48.0	48.0	+0.0 [-7.0,+7.0]
	8	51.0	73.5	**+22.5 [+10.7,+34.3]**	28.5	41.0	**+12.5 [+6.4,+18.6]**	62.5	74.0	**+11.5 [+1.7,+21.3]**	44.0	43.5	-0.5 [-12.5,+11.5]
	10	49.0	72.5	**+23.5 [+12.8,+34.2]**	29.5	43.0	**+13.5 [+6.1,+20.9]**	60.0	78.5	**+18.5 [+10.2,+26.8]**	41.0	47.0	+6.0 [-3.0,+15.0]
	14	52.5	75.5	**+23.0 [+14.7,+31.3]**	30.0	41.0	**+11.0 [+3.3,+18.7]**	64.5	77.5	**+13.0 [+1.7,+24.3]**	33.0	42.0	+9.0 [-0.9,+18.9]
	18	43.0	77.5	**+34.5 [+20.8,+48.2]**	31.0	43.5	**+12.5 [+4.0,+21.0]**	57.5	70.5	**+13.0 [+2.0,+24.0]**	28.0	45.0	**+17.0 [+7.6,+26.4]**
carrot on plate	0	58.5	54.5	-4.0 [-11.7,+3.7]	32.0	28.0	-4.0 [-12.2,+4.2]	58.5	54.5	-4.0 [-11.7,+3.7]	32.0	28.0	-4.0 [-12.1,+4.1]
	4	57.5	52.5	-5.0 [-11.5,+1.5]	41.5	32.0	**-9.5 [-12.1,-6.9]**	56.0	53.0	-3.0 [-10.8,+4.8]	36.5	30.0	-6.5 [-16.5,+3.5]
	8	62.0	53.0	-9.0 [-21.1,+3.1]	42.5	31.0	**-11.5 [-17.1,-5.9]**	55.0	52.5	-2.5 [-10.8,+5.8]	43.0	28.5	**-14.5 [-21.1,-7.9]**
	10	61.5	53.5	**-8.0 [-15.0,-1.0]**	42.5	29.0	**-13.5 [-20.7,-6.3]**	56.0	55.0	-1.0 [-10.7,+8.7]	40.0	28.5	**-11.5 [-18.7,-4.3]**
	14	58.5	51.5	-7.0 [-17.4,+3.4]	32.0	27.5	-4.5 [-9.7,+0.7]	53.5	52.0	-1.5 [-13.8,+10.8]	39.0	37.5	-1.5 [-9.2,+6.2]
	18	56.0	47.0	**-9.0 [-17.7,-0.3]**	32.0	28.5	-3.5 [-10.0,+3.0]	49.5	54.0	+4.5 [-6.9,+15.9]	38.0	41.0	+3.0 [-3.6,+9.6]

CGVD prevents this collapse for both policies. The performance gap also widens as the environment becomes more cluttered: the per-seed slope of the CGVD$-$baseline difference is $+1.54$ pp per distractor for π_0 ($p<0.001$) and $+0.74$ for GR00T ($p=0.02$). A similar though smaller benefit appears under random clutter at high density.

Conversely, on *put carrot on plate* the baseline policies exhibit a slight performance increase at moderate distractor densities before degrading. This likely arises because a moderate amount of contextual clutter better aligns with the object-rich scenes present in their pre-training distributions, providing visual anchors for reasoning.

Here CGVD never significantly outperforms the baseline, and at several densities it significantly underperforms it, most sharply for GR00T (Table I). This suggests that when a task naturally benefits from contextual clutter, aggressively masking the scene deprives the VLA of useful environmental reasoning, and inpainting artifacts can further disrupt the spatial geometry. Therefore, while CGVD provides critical protection against adversarial semantic distractors, the intervention is task- and distractor-type-conditional rather than universally beneficial.

Qualitative attention visualizations consistent with this working hypothesis are provided in the code repository, extracted as the final-query-token attention over the 256 SigLIP visual tokens, averaged across the layers and heads of the PaliGemma backbone. We emphasize that such visualizations are entailed by the intervention itself, since removing a distractor's pixels necessarily removes attention on them, and that no attention statistic is measured in this paper. Alternative explanations, such as CGVD restoring the observation toward the clutter-sparse Bridge training distribution or simply deleting the wrong-grasp option, remain consistent with our data. Indeed, the mean-color-fill ablation in §IV-D, where removing the same semantic content with a cruder fill forfeits most of the gain, indicates that fill appearance rather than distractor semantics alone carries a large share of the effect.

C. Fine-Grained Semantic Grounding

The precision-reasoning gap is most acute when distractors share the target's semantic class but differ in attributes: complex queries like "Put spoon with green handle on towel" are often reduced to a bag-of-words, entangling the target with fully green objects or dropping modifiers. We use the *Attribute* type (§IV-A): a green-handled spoon target among same-category spoons differing in color or material. We evaluate the baseline and CGVD across 0–4 attribute distractors using two prompt structures: Simple ("Put green spoon on towel") and Complex ("Put spoon with green handle on towel"). CGVD receives the same instruction as the policy; its SAM 3 target prompt is the instruction's attribute phrase ("green spoon" / "spoon with green handle"). Coverage note: the attribute type is evaluated only with π_0, one task, 0–4 distractors (vs. Fig. 2's full grid); attribute claims are scoped accordingly.

Table II reveals a two-sided dependence on descriptive density. The baseline degrades only mildly with clutter under the simple prompt, but collapses under the compositional one, dropping from 86.0% at zero distractors to 57.0% at four. This confirms the bag-of-words brittleness. For CGVD, the complex prompt is what unlocks the benefit: because SAM 3 exploits the rich attribute phrase to isolate the correct instance, CGVD holds a stable success floor under the compositional query, and its $+16.0$ pp gain at four distractors is the one delta in this study whose confidence interval excludes zero. Under the simple prompt, by contrast, CGVD provides no reliable benefit, as the terse phrase "green spoon" gives the segmenter too little descriptive signal to separate the target from attribute-conflicting spoons. Both arms of this study were re-executed for this version under the instruction-consistent protocol above. In the earlier version's runs, the segmenter received the scene-informed phrase "spoon with green handle" under both prompt structures, which restores large gains at higher densities but encodes scene knowledge that is not derivable from the simple instruction; removing this asymmetry lowered CGVD's simple-prompt numbers, a correction that ran against our own interest. The five-seed sample matches the study's original protocol, and per-cell values in this table should be read as descriptive. CGVD converts descriptive density into robustness; it does not

TABLE II

ATTRIBUTE DISTRACTOR SENSITIVITY ON *put spoon on towel* (π_0, 100 EPISODES/CELL, 5 MATCHED SEEDS). Δ: 95% CI FROM PER-SEED PAIRED DELTAS; BOLD = CI EXCLUDES ZERO; OTHERWISE WITHIN RUN-TO-RUN NOISE.

	Simple Prompt			Complex Prompt		
# Distractors	π_0	**+CGVD**	**Δ [95% CI]**	π_0	**+CGVD**	**Δ [95% CI]**
0	87.0	85.0	-2.0 [-13.3,+9.3]	86.0	87.0	+1.0 [-5.8,+7.8]
1	79.0	80.0	+1.0 [-7.1,+9.1]	74.0	69.0	-5.0 [-26.9,+16.9]
2	76.0	79.0	+3.0 [-14.9,+20.9]	69.0	77.0	+8.0 [-9.9,+25.9]
3	77.0	75.0	-2.0 [-20.4,+16.4]	64.0	74.0	+10.0 [-7.6,+27.6]
4	75.0	70.0	-5.0 [-24.6,+14.6]	57.0	73.0	**+16.0 [+1.2,+30.8]**

TABLE III

ABLATION STUDY (π_0, *put spoon on towel*, 18 SEMANTIC DISTRACTORS; 200 EPISODES/ROW). ROWS REMOVE OR SUBSTITUTE ONE COMPONENT. SR: MEAN $\pm$ SD OVER PER-SEED RATES; Δ VS. FULL PIPELINE WITH 95% CI FROM PAIRED DELTAS. BASELINE/FULL ROWS REUSE THE FIG. 2 RUNS; THE ROBOT-MASK EFFECT IS NOT DISTINGUISHABLE FROM ZERO.

Configuration	**SR (%)**	**Δ [95% CI]**
Baseline (no CGVD)	43.0 ± 16.0	-34.5 $[-48.2, -20.8]$
CGVD (full pipeline)	$\mathbf{77.5} \pm 6.3$	—
LaMa $\rightarrow$ mean-color fill	56.5 ± 16.0	-21.0 $[-35.3, -6.7]$
– Two-layer target refinement	65.0 ± 8.8	-12.5 $[-20.3, -4.7]$
– Robot mask protection	73.0 ± 7.1	-4.5 $[-10.4, +1.4]$

rescue a prompt that lacks it.

D. Ablation Studies

We ablate the pipeline at a single operating point—*put spoon on towel*, π_0, 18 semantic distractors (Table III); π_0's higher success rates provide headroom to resolve component contributions. The attributions are claims about this operating point only: §IV-B shows the pipeline's net effect reverses in other regimes.

Removing the Two-Layer Target Refinement reduces SR from 77.5% to 65.0%. Without cross-validation, genuine targets are erroneously inpainted out of the scene, meaning CGVD's own protection fails, which is why we describe it as conditional (§III-A). To quantify the residual failure rate of the full pipeline, we audited a fresh 200-episode re-execution of the headline condition against the renderer's ground-truth target mask. The inpainting mask overlapped the target at $t=0$ in 4 of the 200 episodes (2.0%), and in all four cases the target was erased essentially completely. The protection is therefore strong but probabilistic in practice, differing from sensitivity-probe approaches in failure rate rather than failure mode. This re-execution also replicates the headline cell, at 76.0% against the original 77.5%. Substituting a mean-color fill for LaMa incurs the largest single drop, to 56.5%, despite identical masks and removed content; we hypothesize, without measuring it, that the stark region boundaries act like out-of-distribution patches to the ViT backbone. Notably, a majority of the gain at this operating point thus depends on how the removed regions are filled rather than on the removal alone. Finally, removing Robot Mask Protection reduces SR to 73.0%. This -4.5 pp effect is not statistically distinguishable from zero at our sample size, so we treat robot-mask protection as qualitatively motivated—without it, the compositing mask occasionally and visibly occludes the robot arm—rather than as a measured gain.

Deployable robot mask. To measure the cost of the renderer mask's simulation privilege, we additionally ran the full pipeline with a per-frame SAM 3 prediction of the robot mask, on the same seeds. No accuracy difference was detected: success reaches 74.0% against 76.0% for a ground-truth-mask re-execution (paired $\Delta = -2.0$ pp), though at this sample size the experiment only excludes accuracy costs larger than about 11 pp. The costs lie in latency and reliability instead. The deployable path adds 240.7 ± 1.9 ms per frame, roughly 0.96 s per 4-step chunk, and SAM 3 returns no robot detection at all in 20.1% of the 12,400 audited frames, falling back to the previous mask and leaving the arm's leading edge unprotected over formerly inpainted regions. Forward-kinematics projection (§III-F), which avoids this failure class entirely, remains the sensible hardware path.

Vocabulary sensitivity. Because $\mathscr{D}$ coincides with the injected inventory (§III-A), we re-ran the headline condition on the same seeds with two frozen vocabularies to measure how much of the gain depends on that privilege. With a vocabulary that misses the injected clutter ($\mathscr{D} = \{$ladle, whisk, bowl, cup$\}$), success collapses to 45.5%—statistically indistinguishable from the 43.0% baseline (paired Δ vs. baseline $+2.5$, 95% CI $[-6.6, +11.6]$). With a *generic* 12-category tabletop vocabulary not derived from the injection list (fork, knife, spatula, scissors, ladle, whisk, plate, bowl, cup, mug, pan, pot), success is 76.5%—statistically indistinguishable from the oracle vocabulary (Δ vs. oracle re-execution $+0.5$ $[-4.7, +5.7]$; vs. baseline $+33.5$ $[+22.3, +44.7]$). CGVD's benefit therefore requires vocabulary coverage of the clutter, but not knowledge of the exact inventory. We caution that the miss condition is not a no-op: the full pipeline still runs, the scene is still edited (mask coverage falls to roughly 2% from 14%), the cache still freezes, and the erasure and blending risks remain live. Matching the baseline here means the benefit is forfeited, not that the intervention is absent. Usefully, a near-empty distractor mask is itself a one-line vocabulary-miss detector.

E. Comparison with a Run-Time Intervention Baseline

We compare CGVD against BYOVLA [9], the closest prior run-time observation intervention, at the headline condition (same 10 matched seeds, 200 episodes), reimplementing it from its reference code with three substitutions chosen to remove confounds: GPT-4o replaced by the *same* closed vocabulary $\mathscr{D}$ CGVD receives, GroundingDINO+SAM2 by SAM 3 (shared with CGVD), and Octo's fixed-key stochastic probe by fixed-seed π_0 forward passes. The mechanism is kept faithful,[2] and the probe demonstrably discriminates here (mean 2.0 of 32.1 candidate regions flagged per chunk, range 0–16).

[2]Per-object Gaussian-blur perturbation (random kernel 15–30); translation-only action deltas with the reference threshold (0.002 m, same physical units); sensitive-region inpainting with the *shared* LaMa inpainter (dilation 10); intervention before every 4-step action chunk (the π_0 Bridge configuration; also the reference cadence).

BYOVLA reaches 32.5%, numerically below the 43.0% baseline ($\Delta = -10.5$, within run-to-run noise) and far below CGVD's 77.5% (paired $\Delta = -45.0$, 95% CI $[-56.4, -33.6]$). Its cost is also substantial: each intervention takes 11.3 ± 2.0 s, dominated by roughly 33 policy forward passes, which per 4-step action chunk amounts to 11.6 s against CGVD's 0.38 s, an overhead of about 30×. We attribute this to two causes. Per-chunk re-segmentation and re-inpainting makes the background flicker across chunks, precisely the fill-appearance instability that the mean-color-fill row of Table III shows the policy is sensitive to, and that CGVD's cached clean plate avoids. In addition, the conservative sensitivity threshold leaves most of the 18 distractors in view. Two caveats apply: this is our reimplementation rather than the authors' system, and eighteen dense distractors lie well outside BYOVLA's original operating regime of a real robot with a handful of distractors; we also did not tune its threshold. The fair conclusion is that per-chunk probe-and-inpaint does not scale to dense clutter, not that the method fails in its intended regime.

F. Latency Analysis

CGVD concentrates its expensive operations (SAM 3, LaMa) on the initialization frame; for $t > 0$ the system only composites against the cached clean scene. All timings below were measured on an NVIDIA A10G (23 GB) across the evaluation runs, as means ± SD over episodes; an earlier version of this paper reported a single blended per-step figure, which we replace here with decomposed timings. The policy forward pass is unaffected: 318.1 ± 6.6 ms baseline (500 episodes) versus 318.5 ± 5.5 ms under CGVD (1,100 episodes), with one forward per 4-step action chunk. Per-frame compositing adds 15.6 ± 0.4 ms, or 62 ms per chunk, which amounts to roughly 10% additional episode wall time excluding initialization ($9.11 \rightarrow 10.00$ s in the attribute runs). Initialization is a one-off per-episode cost that grows with the number of segmentation prompts, from 0.68 ± 0.29 s in the attribute scenes to 0.91 ± 0.02 s in the 18-distractor scenes (measured directly over warm episodes; the first episode of a session additionally pays one-time model loading), delaying each episode's first action. The overhead is modest for the quasi-static tabletop tasks evaluated here, though the startup delay remains a real cost for time-critical settings.

V. Limitations

CGVD's validity envelope is defined by the following assumptions and scope limits.

Static scene. The clean scene is cached at $t{=}0$ and α is fixed (§III-F). A moved distractor desynchronizes the cache; worse, an object *entering* the workspace after $t{=}0$ can be blended toward the cached background and hidden from the policy while physically present—a hazard, not merely an accuracy loss, outside access-controlled workspaces. Continuous re-segmentation would lift the assumption but is prohibitive at control rate.

Contextually informative clutter. On *put carrot on plate* CGVD never outperforms the baseline and significantly underperforms it at several densities (up to -14.5 pp for GR00T; §IV-B): the intervention is task- and distractor-type-conditional.

Simulation-only evaluation and privileged inputs. No real-robot experiment is reported, and two inputs are simulation-privileged: the renderer robot mask and the per-episode distractor inventory $\mathscr{D}$. SAM 3 recall over $\mathscr{D}$ and LaMa fill fidelity on real textured scenes (the largest single factor in Table III) are unvalidated on hardware; the per-frame SAM 3 robot mask is measured only in simulation (§IV-D), and FK projection on hardware is untested.

Single target instance. Layer 2 keeps only the top-scoring component (§III-C); a scene with two genuine target instances would lose one to inpainting.

Occlusion fabrication and in-flight blending. Whatever the arm occluded at $t{=}0$ is synthesized, not observed, in the cached scene (Eq. 6); and since α protects the target only at its $t{=}0$ location, the *moving* target crosses previously inpainted regions during execution: in the 200-episode headline re-execution, its ground-truth mask overlapped the fixed inpaint mask by more than 10% of its pixels at some frame in 35% of episodes, exposing the grasped object to partial blending.

VI. Conclusion

We introduced CGVD, a training-free inference framework that mitigates the Precision-Reasoning Gap via language-grounded segmentation and inpainting, without architectural modification. Across more than 20,000 SimplerEnv episodes with π_0 and GR00T, CGVD prevents performance collapse under dense semantic clutter and, in an exploratory compositional-prompt condition, improves attribute adherence, relative to the un-augmented policies; at the headline condition it outperforms a vocabulary-matched BYOVLA reimplementation by 45 pp at roughly a thirtieth of its per-chunk cost, in a regime outside its design envelope and with its threshold untuned. The benefit is conditional: where contextual clutter aids the policy, CGVD never outperforms the baseline and often significantly underperforms it. Bounded by static-background and closed-vocabulary assumptions, these results position inference-time visual distillation as a practical defense against semantically confusable clutter. Future work will explore real-time mask updating, intrusion detection over the compositing regions, open-vocabulary distractor sets, and criteria for when to distill.

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